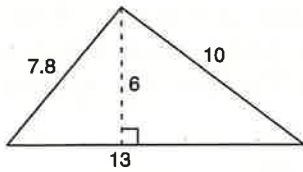


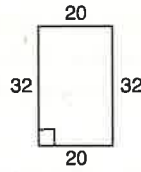
Practice 4

A. Find the area and perimeter of each figure.

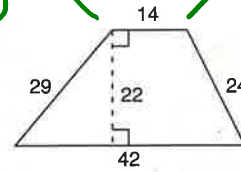
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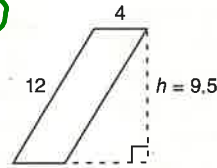
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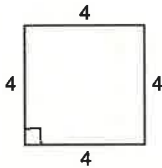
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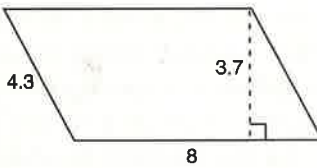
6.



2.



4.



trapezoid

$$A = \frac{1}{2}h(b_1 + b_2)$$

$$= \frac{1}{2}(22)(42 + 14)$$

$$= (11)(56)$$

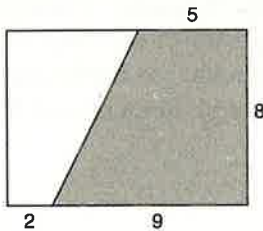
$$= 616 \text{ units}^2$$

$$P = 14 + 24 + 42 + 29$$

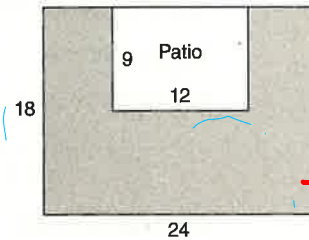
$$= 109 \text{ units}$$

B. Choose the one best answer to each question.

Question 7 refers to the following figure.



9. Martin is building a rectangular patio centered on one side of his yard. The rest of his yard, shown in the diagram, is planted in grass.



$$A = b \times h$$

$$= 4 \times 9.5$$

$$= 38 \text{ units}^2$$

$$P = 4 + 4 + 12 + 12$$

$$= 32 \text{ units}$$

7. What is the area in square inches of the shaded portion of the rectangle?

- A. 38
B. 40
C. 56
D. 88

8. The four sides of a rectangle measure 9 feet, 6 feet, 9 feet, and 6 feet. What is the area of the rectangle in square feet?

- A. 30
B. 36
C. 54
D. 81

If the measurements in the diagram are in feet, what is the square footage of the grass portion of Martin's yard?

- A. 108
B. 162
C. 324
D. 432

10. A square measures 6 centimeters on one side. What is the perimeter of the square in centimeters?

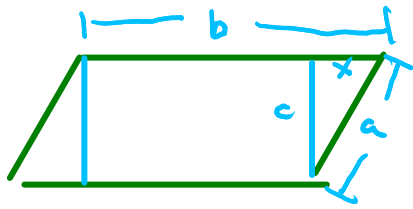
- A. 12
B. 24
C. 36
D. 216

$$A = (24 \times 18) - (9 \times 12)$$

$$= 432 - 108$$

$$= 324 \text{ ft}^2$$

Answers and explanations begin on page 683.



$$a^2 = c^2 + x^2$$

Lesson 5: Circles

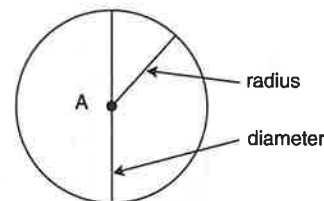
Key Ideas

- Pi, expressed as 3.14, is the ratio of the circumference of a circle to its diameter.
- To find circumference, multiply pi by the diameter (distance across) the circle.
- The radius is the distance from the center to the edge of the circle. To find area, multiply pi by the square of the radius.

Circumference and Area

A **circle** is a closed set of points that are all the same distance from a single point, the center of the circle. The **circumference** of a circle is its perimeter, or the distance around the circle. The **area** of a circle is the space inside the circle.

To find perimeter and area of a circle, you need to know two other measures of a circle. The **diameter** is a line segment with endpoints on the circle that passes through the center of the circle. The **radius** is a line segment that connects the center of the circle to any point on the circle. As you can see from the diagram, the radius is one-half the diameter.



The formulas for circumference and area use a special quantity called **pi** (π). Pi is the ratio of the circumference to the diameter. It is equal to approximately 3.14. As far as we know, the digits for pi continue infinitely, so calculations with pi are always approximations. For the *Mathematical Reasoning Test*, you will use 3.14 as the value of pi. Below is the formula for finding the circumference of a circle.

Circumference = $\pi \times \text{diameter}$, or $C = \pi d$

Example 1: A china plate has a gold rim. If the plate's diameter is 10.5 inches, what is the distance around the rim to the nearest tenth of an inch?

Use the formula: $C = \pi d$
 $= 3.14(10.5)$
 $= 32.97$, which rounds to **33.0 inches**

Use this formula to find the area of a circle: $\text{Area} = \pi \times \text{radius}^2$, or $A = \pi r^2$.

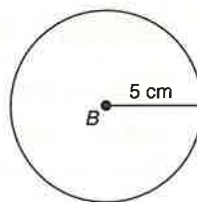
Example 2: The circular surface of a satellite component must be covered with heat-resistant tiles. If the radius of the component is 4 meters, what is the area in square meters?

Use the formula: $A = \pi r^2$
 $= 3.14(4^2)$
 $= 3.14(16)$
 $= 50.24$ **square meters**

In some situations, you may need to solve for either the diameter or radius. Remember, the diameter is twice the radius ($d = 2r$), and the radius is one-half the diameter: $r = \frac{1}{2}d$.

Example 3: What is the circumference of circle B to the nearest tenth of a centimeter?

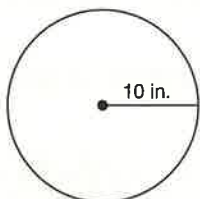
1. The radius of the circle is 5 cm. Therefore, the diameter is 2×5 , or 10 cm.
2. Use the formula: $C = \pi d$
 $= 3.14(10)$
 $= 31.4 \text{ cm}$



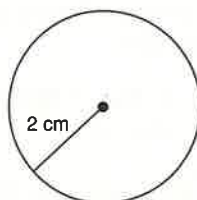
Practice 5

A. Find the circumference and area of each circle. Round answers to the nearest tenth.

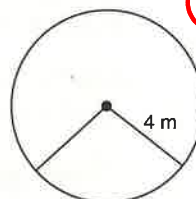
1.



2.



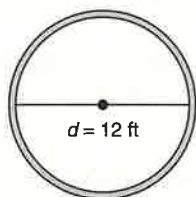
3.



① $C = 2\pi r$
 $= 2\pi(10)$

B. Choose the one best answer to each question.

Questions 4 and 5 refer to the following drawing.



4. If workers lay a tile border around the edge of the fountain shown in the diagram, how many feet long will the border be to the nearest foot?

- A. 19
- B. 36
- C. 38
- D. 57

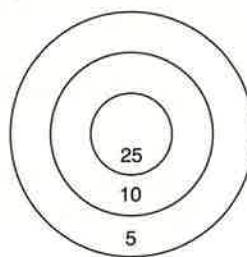
5. Which of the following expressions could be used to find the area of the bottom surface of the fountain?

- A. 3.14×6
- B. 3.14×6^2
- C. 3.14×12
- D. 3.14×12^2

6. The radius of a circle is 6.5 cm. What is the diameter of the circle in centimeters?

- A. 3.25
- B. 13.0
- C. 33.16625
- D. 132.665

7. On the target below, the 5- and 10-point bands are each 2 inches wide, and the 25-point inner circle has a diameter of 2 inches. (Note: figure not drawn to scale.)



To the nearest inch, what is the outer circumference of the 10-point band?

- A. 6
- B. 13
- C. 19
- D. 113

Answers and explanations begin on page 683.

Lesson 6: Volume

Key Ideas

- Volume is measured in cubic units that may be written using an exponent: 6 cubic inches or 6 in^3 .
- Find volume by multiplying the area of one base by the height of the object.
- You can also use formulas to find volume.

Rectangular Solids, Cubes, Cylinders, and Spheres

Volume, also called **capacity**, is the measure of space inside a three-dimensional object. You measure volume in cubic units. In other words, if the sides of an object are measured in inches, the volume is the number of cubes (one inch per side) you would need to fill the object.

Many common three-dimensional objects have at least two identical and parallel faces. Think of a cereal box or a soup can. Both objects have identical faces at the top and bottom of the container. Either of these faces can be called the base of the object. To find the volume of any container with identical bases, multiply the area of one base by the height of the object: $\text{Volume} = \text{area of base} \times \text{height}$.

Another way to find the volume of an object is to use the formula that applies specifically to that object.

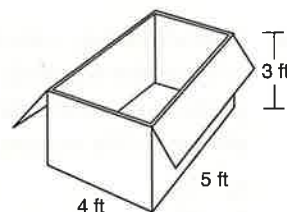
rectangular prism	Volume = base $\times$ height	$V = Bh$
cube	Volume = edge ³	$V = e^3$
cylinder	Volume = pi $\times$ radius ² $\times$ height	$V = \pi r^2 h$
sphere	Volume = $\frac{4}{3} \times \text{pi} \times \text{radius}^3$	$V = \frac{4}{3} \pi r^3$

In the examples below, formulas are used to find the answers, but the problems can also be solved by simply multiplying the area of the base by the height.

A **rectangular prism** has two identical rectangular bases. The remaining sides of the figure are also rectangles.

Example 1: A cardboard box has the dimensions shown in the diagram. What is the volume of the box in cubic feet?

Use the formula: $V = Bh$, where B is length $\times$ width
 $= (4 \times 5)(3) = \mathbf{60 \text{ cubic feet}}$



A **cube** is a rectangular prism with six identical faces. In a cube, each edge (where the sides meet) is the same length.

Example 2: A wood block measures 2 inches per edge. What is the volume of the block?

Use the formula: $V = e^3$
 $= 2^3 = \mathbf{8 \text{ cubic feet}}$



A **cylinder** has two circular bases. The bases are connected by a curved surface. Cans, barrels, and tanks are often in the shape of cylinders.

GED® TEST TIP

Learn which area formula goes with which shape. Once you know the area formulas, you can change many of them to volume by putting an h at the end. Compare:
Area of circle: πr^2
Volume of cylinder: $\pi r^2 h$

Example 3: A storage tank has a radius of 1.5 meters and a height of 3 meters. What is the volume of the tank to the nearest cubic meter?

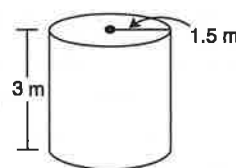
Use the formula:

$$V = \pi r^2 h$$

$$= 3.14(1.5^2)(3) = 21.195 \text{ m}^3,$$

which rounds to 21 cubic meters.

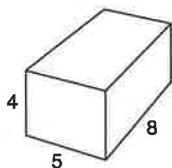
or m^3



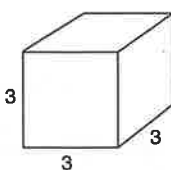
Practice 6.1

A. Find the volume of each object to the nearest whole unit.

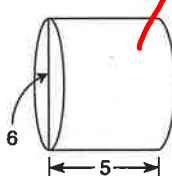
1.



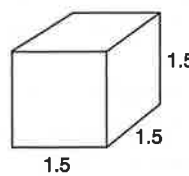
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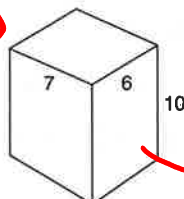
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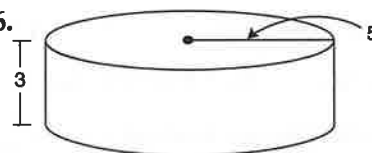
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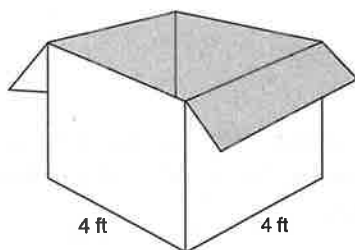


6.



B. Choose the one best answer to each question.

Question 7 refers to the following drawing.

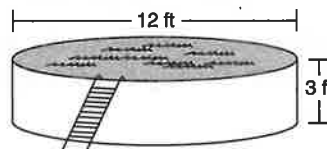


7. A rectangular box with a volume of 80 cubic feet has the length and width shown in the drawing. What is the height, in feet, of the box?

- A. 5
- B. 10
- C. 16
- D. 20

Question 9 refers to the following drawing.

9. Linda adds a water stabilizer to her children's swimming pool once a week. The instructions tell her to add one scoop of the product for every 20 cubic feet of water. About how many scoops should she add per week?



- A. 3
- B. 6
- C. 17
- D. 36

Answers and explanations begin on page 683.

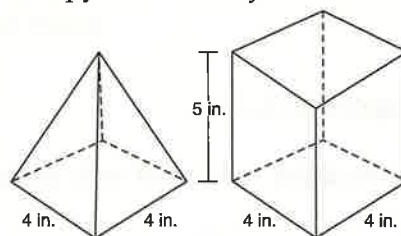
8. A wooden crate measures 5 feet along each edge. What is the crate's volume in cubic feet?

- A. 15
- B. 25
- C. 125
- D. 150

Volume of Pyramids and Cones

A **pyramid** is a three-dimensional object with four triangle faces that connect to the same vertex. The base of a pyramid can be any closed figure, but the pyramids that you will see on the GED® *Mathematical Reasoning Test* will all have square bases.

The rectangular prism and the pyramid shown here both have identical square bases and the same height. Compare the two figures. As you can see, the pyramid holds much less than the rectangular prism. In fact, it holds only one-third of the rectangular prism's volume.

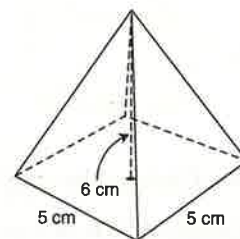


The formula for finding the volume of a pyramid is as written below. Notice that B equals the area of the square base. The height of a pyramid is the perpendicular distance from the base to the vertex at the top.

square pyramid Volume = $\frac{1}{3} \times \text{base} \times \text{height}$

Example 1: Find the volume of the pyramid shown below.

The length of a base edge is 5 cm. The height of the pyramid is 6 cm. Ignore the diagonal edges.



$$\begin{aligned} \text{Apply the formula: } V &= \frac{1}{3} Bh \\ &= \frac{1}{3} (5^2)(6) \\ &= \frac{1}{3} (25)(6) \\ &= 50 \text{ cm}^3 \end{aligned}$$

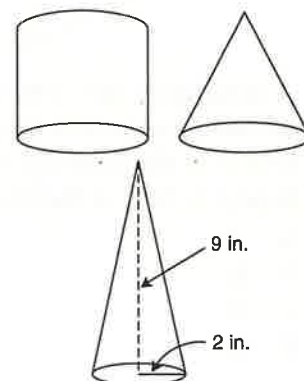
A **cone** is similar to a cylinder. Both have a circular base and a curved side. The curved side of a cone slants inward so that it meets at a point, or vertex. The volume of a cone is $\frac{1}{3}$ of the volume of a cylinder with the same size base and height.

On the GED® Formula Sheet, notice that the formula for the volume of a cone contains the formula for finding the area of a circle (πr^2).

cone Volume = $\frac{1}{3} \times \pi \times \text{radius}^2 \times \text{height}$

Example 2: Find the volume of the cone shown below.

The radius of the base is 2 inches, and the height is 9 inches.



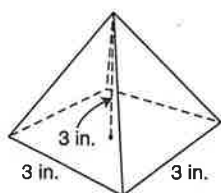
$$\begin{aligned} \text{Apply the formula: } V &= \frac{1}{3} \pi r^2 h \\ &= \frac{1}{3} (3.14)(2^2)(9) \\ &= \frac{1}{3} (3.14)(4)(9) \\ &= 37.68 \text{ in.}^3 \end{aligned}$$

NOTE: In volume formulas for cones and square pyramids, the factor $\frac{1}{3}$ is shown first. Don't begin with this factor. Multiply the other factors first and then divide by 3, unless one of the numbers is easily divided by 3.

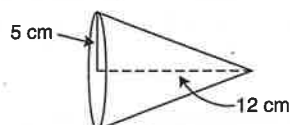
Practice 6.2

A. Find the volume of each object to the nearest unit.

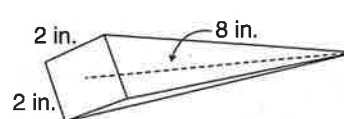
1.



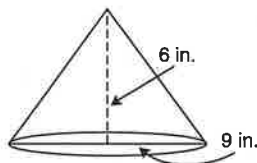
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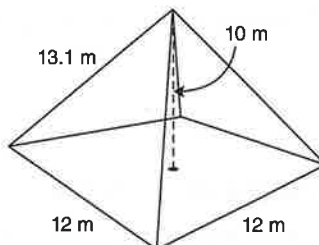
5.



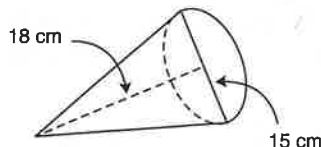
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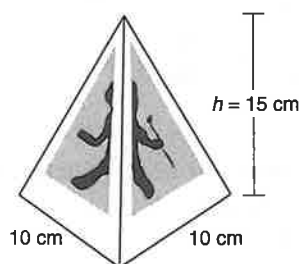


6.



B. Choose the one best answer to each question.

7. Advertisers have designed this pyramid-shaped package to hold action figures.



9. Which of the following is a true statement about the figures shown below?

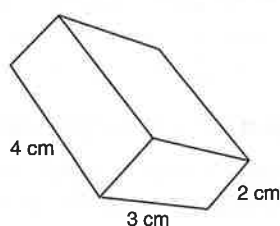


Figure A

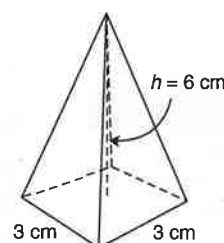


Figure B

After testing the design, the manufacturer decides to increase both the length and width of the base by 4 cm. How many more cubic centimeters will the new package hold than the package shown above?

- A. 480
 - B. 500
 - C. 980
 - D. 2,940
8. The height of a cone is half the diameter of its base. If the cone's height is 4 inches, what is the cone's volume to the nearest cubic inch?
- A. 21
 - B. 48
 - C. 67
 - D. 268

- A. The volume of figure A equals the volume of figure B.
- B. The volume of figure B is greater than the volume of figure A.
- C. Both figures A and B have a volume greater than 20 cubic centimeters.
- D. The volume of figure B is less than the volume of figure A.

10. A cone's base is a circle with a radius of 8 inches. The cone's height is 15 inches. What is the cone's approximate volume in cubic inches?

- A. 125
- B. 250
- C. 500
- D. 1,000

Answers and explanations begin on page 683.

Lesson 7: Surface Area

Key Ideas

- The surface area is the area of the total surface of a three-dimensional object.
- To find the surface area of a cube, you need to multiply the area of one side by 6.
- To find the surface area of a rectangular prism and a pyramid, you need to use formulas to find several dimensions and combine them.

GED® TEST TIP

Memorize the process for finding the surface area of a cube. The formulas for surface areas of prisms and pyramids are listed on the online Formula Sheet on the GED® Mathematical Reasoning Test.

The **surface area** is the total area of the outside **faces** of three-dimensional figures. This is different from the **volume**, which is the capacity of what a figure can hold. Surface area is expressed in square units.

Surface Area of a Square Prism

The surface area of a **square prism (cube)** is the sum of the areas of the six squares that form the prism. The area of one square is s^2 . Since these sides are the same, find the area of one side and multiply by 6.

Example 1: Find the surface area of a cube with sides of 3 inches each.

$$SA = 6s^2$$

$$SA = 6(3 \text{ in})^2 = 6(9 \text{ in}^2) = 54 \text{ in}^2$$

Surface Area of a Rectangular Prism

The surface area of a **rectangular prism** is the sum of the areas of the six rectangles that form the prism.

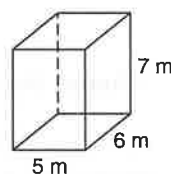
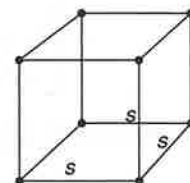
Example 2: Find the surface area of this box.

$$SA = ph + 2B$$

$$SA = \text{perimeter} \times \text{height} + 2(\text{area of base})$$

$$SA = (5 \text{ m} + 6 \text{ m} + 5 \text{ m} + 6 \text{ m}) \times 7 \text{ m} + 2(6 \text{ m} \times 5 \text{ m})$$

$$SA = (22 \text{ m})7 \text{ m} + 2(30 \text{ m}^2) = 154 \text{ m}^2 + 60 \text{ m}^2 = 214 \text{ m}^2$$



Surface Area of a Pyramid

The surface area of a **pyramid** is $\frac{1}{2}$ the perimeter of the base (p) times the **slant height** (s) plus the area of the base (B). The slant height is the length of a line segment from the **apex** (top) to the **base**.

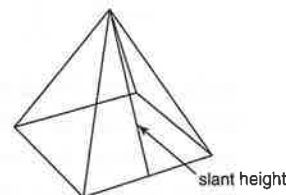
Example 3: What is the approximate surface area of a pyramid with a slant height of 5 feet, a base with a perimeter of 14 feet, and a base with an approximate area of 12 square feet?

$$SA = \frac{1}{2} ps + B$$

$$SA = \frac{1}{2} \text{perimeter} \times \text{slant height} + \text{base area}$$

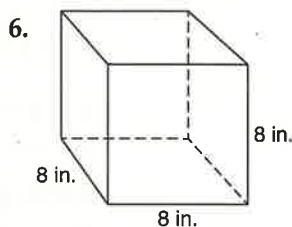
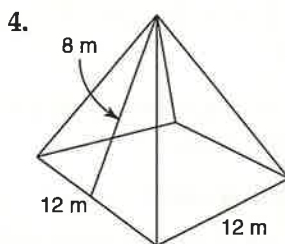
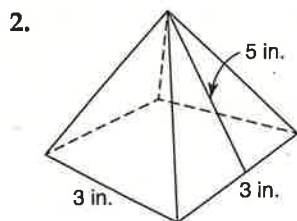
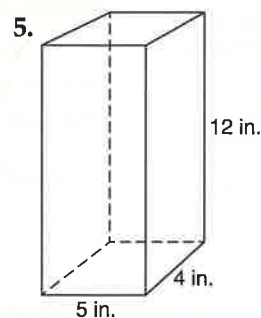
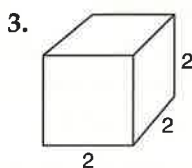
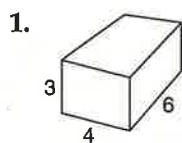
$$SA = \frac{1}{2}(14 \text{ ft})(5 \text{ ft}) + 12 \text{ ft}^2$$

$$SA = \frac{1}{2}(70 \text{ ft}^2) + 12 \text{ ft}^2 = 35 \text{ ft}^2 + 12 \text{ ft}^2 = 47 \text{ ft}^2$$



Practice 7.1

A. Find the surface area of each object in square units. Use the formulas on page 402.



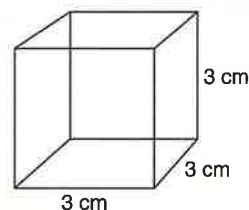
B. Choose the one best answer to each question. Use the formulas on page 402.

7. All the edges of a metal box are of equal length. If the surface area is 150 square inches, what is the length, in inches, of each edge of the box?

A. 5
B. 6
C. 25
D. 50

9. If the dimensions of the box below are doubled, by how many square centimeters does the surface area increase?

A. 8
B. 54
C. 162
D. 216



8. Which of the following is a true statement about the figures shown below? (Measurements indicated are all in the same units. Note that the figures are not drawn to scale.)

Answers and explanations begin on page 684.

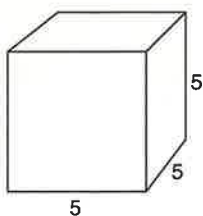


Figure A

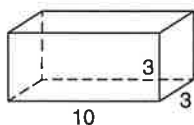


Figure B

- A. The surface area of Figure A is equal to the surface area of Figure B.
B. The surface area of Figure A is half of the surface area of Figure B.
C. The surface area of Figure A is greater than the surface area of Figure B.
D. The sum of the surface areas of both Figure A and Figure B is greater than 300.

Key Ideas

- The formulas for surface area of cylinders, cones, and spheres all use dimensions of a circle.
- All of the formulas use π and the radius of a circle.

GED® TEST TIP

The three formulas on this page are available on the online Formula Sheet on the GED® Mathematical Reasoning Test.

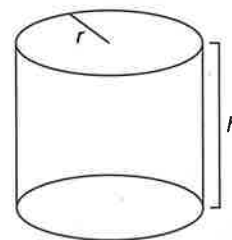
Finding the surface areas of cylinders, cones, and spheres involves using some dimensions of circles.

Surface Area of a Cylinder

To find the surface area of a cylinder, multiply the circumference ($2\pi r$) by the **height** (h) and add the product to the area of the two ends of the cylinder, each of which is the area of a circle (πr^2).

Example 1: Find the surface area of a cylinder with a radius (r) of 4 inches and a height (h) of 3 inches.

$$\begin{aligned} SA &= 2\pi rh + 2\pi r^2 \\ SA &= (2 \times 3.14 \times 4 \times 3) + (2 \times 3.14 \times 4^2) \\ SA &= (6.28 \times 12) + (6.28 \times 16) = 75.36 \text{ in}^2 + 100.48 \text{ in}^2 \\ &= 175.84 \text{ in}^2 \end{aligned}$$



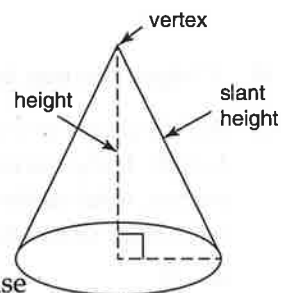
You may find the on-screen calculator helpful to complete these calculations effectively.

Surface Area of a Cone

The surface area of a **cone** is found by combining the **lateral side surface** (πrs) with the area of its **base** (πr^2).

Example 2: Find the surface area of a cone with a radius of 2 feet and a slant height of 5 feet.

$$\begin{aligned} SA &= \pi rs + \pi r^2 \\ SA &= \text{lateral side surface with slant height} + \text{area of base} \\ SA &= \pi(2)(5) + \pi(2^2) \\ SA &= 10\pi \text{ ft}^2 + 4\pi \text{ ft}^2 = 14\pi \text{ ft}^2 \approx 14 \text{ ft}^2 \times 3.14 = 43.96 \text{ ft}^2 \end{aligned}$$

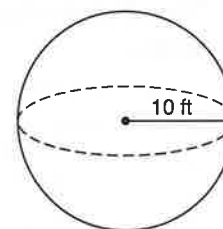


Surface Area of a Sphere

The surface area of a **sphere** is four times the area of a circle with the same radius as the sphere.

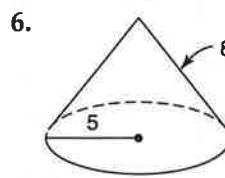
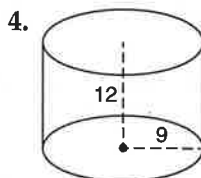
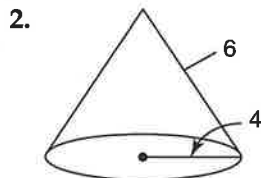
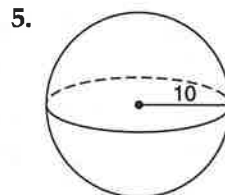
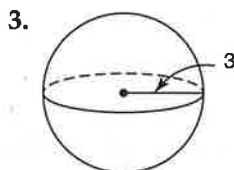
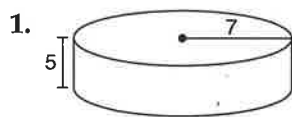
Example 3: What is the surface area of this sphere with a radius of 10 feet?

$$\begin{aligned} SA &= 4\pi r^2 \\ SA &\approx 4 \times 3.14 \times 10^2 \approx 1,256 \text{ ft}^2 \end{aligned}$$



Practice 7.2

A. Find the surface area of each object in square units. Express your answer in terms of π (for example, 36π). Use the formulas on page 404.

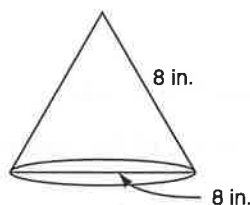


B. Choose the one best answer to each question. Use the formulas on page 404.

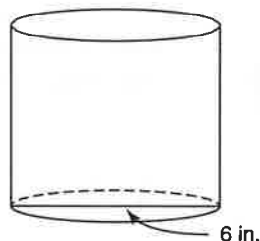
7. If the length of the diameter of a sphere is 8, how many square units is its surface area?

A. 16π
B. 64π
C. 256π
D. 512π

8. A cone, pictured below, has a slant height of 8 inches and a diameter of 8 inches. If the cone's slant height is doubled, and the diameter is halved, which of the following statements must be true?



9. In the cylinder below, the diameter of the circular base is equal to the height of the cylinder. What is the surface area of the cylinder, to the nearest square inch?



A. 57
B. 113
C. 170
D. 339

Answers and explanations begin on page 684.

- A. The new cone will have the same surface area as the original cone.
B. The new cone will have a surface area exactly half of the surface area of the original cone.
C. The new cone will have a surface area exactly double that of the surface area of the new cone.
D. The new cone will have a surface area less than the surface area of the original cone.

Lesson 8: Combined Figures

Breaking Combined Figures into Parts

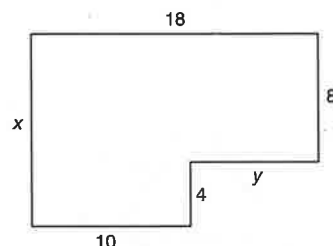
A combined figure puts together geometric figures to form a new shape. To find the perimeter of a combined figure, simply add the lengths of the sides. You may need to solve for one or more missing lengths.

Key Ideas

- Combined figures are made from two or more simple figures.
- To find the perimeter of a combined figure, add the lengths of all the sides.
- To find area or volume of a combined figure, break the figure into parts and find the area or volume of each part; then combine the results.

Example 1: A family room has the dimensions shown in the diagram. All measures are in feet. What is the perimeter of the room?

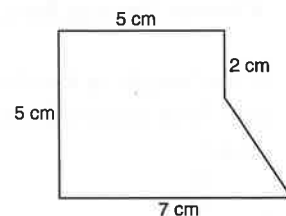
1. Find the missing measures. Measurement x equals the combined lengths of the two opposite walls: $x = 8 + 4 = 12$ ft. You also know that $18 - 10 = 8$, so $y = 8$ ft.
2. Add all distances to find the perimeter:
 $12 + 18 + 8 + 8 + 4 + 10 = 60$ ft.



To find the area or volume of a combined figure, break the figure into parts. Then apply the correct formula to each part.

Example 2: What is the area of the figure in square centimeters?

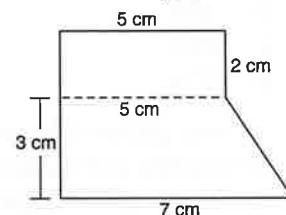
1. Divide the figure into two shapes and find any missing measurements. Here the figure is divided into a trapezoid and a rectangle.



2. Calculate the area of each shape.

$$\begin{aligned}\text{Rectangle: } A &= lw \\ &= 2(5) = 10 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Trapezoid: } A &= \frac{1}{2}h(b_1 + b_2) \\ &= \frac{1}{2}(3)(5 + 7) = 18 \text{ cm}^2\end{aligned}$$

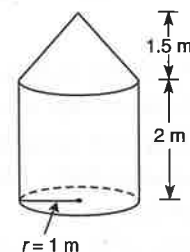


3. Combine: $10 + 18 = 28 \text{ cm}^2$.

Example 3: Find the volume of the container shown below.

Break the figure into a cylinder and a cone and find the volume of each.

1. Cylinder: $V = \pi r^2 h$
 $\approx (3.14)(1^2)(2) = 6.28 \text{ m}^3$
2. Cone: $V = \frac{1}{3}\pi r^2 h$
 $\approx \frac{1}{3}(3.14)(1^2)(1.5) = 1.57 \text{ m}^3$
3. Combine: $6.28 + 1.57 = 7.85 \text{ m}^3$ or 7.85 cu m .



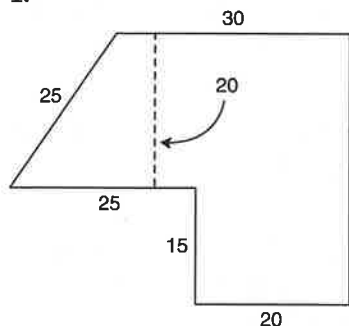
GED® TEST TIP

There is usually more than one way to break down a combined figure. Solve the problem the way that seems easiest to you. The answer will be the same.

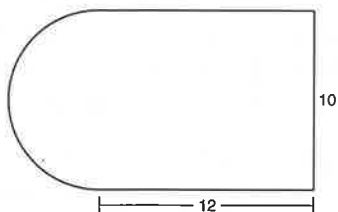
Practice 8

A. Find the perimeter and area of each figure.

1.

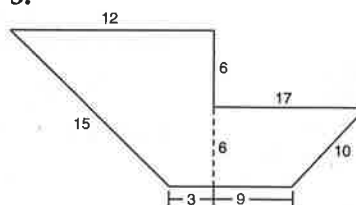


2.



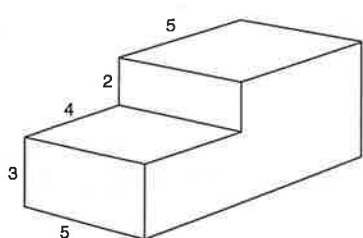
(Hint: Think of the figure as a rectangle and a half circle.)

3.

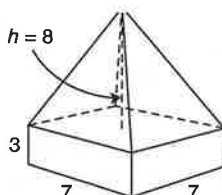


B. Find the volume of each figure to the nearest cubic unit.

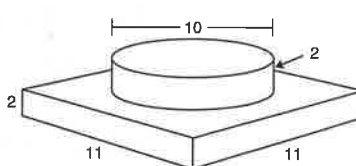
4.



5.

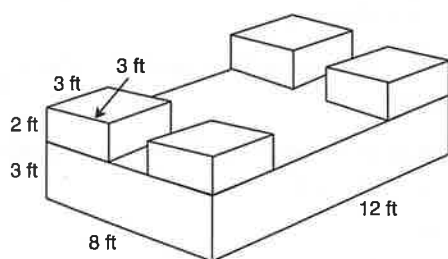


6.



C. Choose the one best answer to each question.

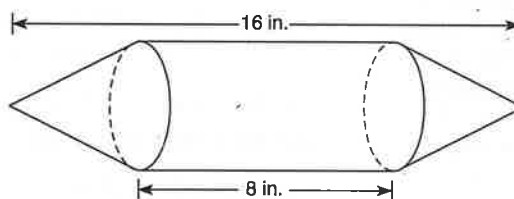
7. A slab of concrete will have four concrete blocks in each corner as shown in the drawing below.



If each corner block has the same dimensions, what is the volume of the structure in cubic feet?

- A. 72
- B. 168
- C. 288
- D. 360

8. A candy package is in the shape of a cylinder with a cone on each end.



If the radius of the cylinder is 2 inches and the cones are identical, what is the capacity of the container to the nearest cubic inch?

- A. 33
- B. 100
- C. 134
- D. 201

Answers and explanations begin on page 684.

Lesson 9: Geometry Calculator Skills

Key Ideas

- When a problem involves many calculations, use the order of operations to decide which operation to do first.
- Check your work by estimating an answer or by quickly repeating the keystrokes to see if the result is the same.

When solving problems with formulas, you can generally save time by performing some or all of the operations on a calculator. Remember the order of operations when solving formulas and always check your work, either by reentering the key sequence or by estimating an answer and comparing your answer to the estimate.

Example 1: A pyramid has a height of 81 feet. The base is in the shape of a square with each side measuring 40 feet. What is the volume in cubic feet of the pyramid?

Use the formula for the volume of a pyramid: $\frac{1}{3}Bh$. Begin by calculating the area of the base: 40×40 , or 40^2 . Multiply that by the height, 81. In the last step, dividing by 3 is the same as multiplying by $\frac{1}{3}$.

Evaluate the exponent.	$40 \text{ } x^2$
Multiply by 81.	$\times 81$
Divide by 3.	$\div 3 \text{ } \text{enter}$
The right side of the display shows the correct answer, 43,200.	43200

To enter this formula using a fraction using one series of keystrokes on the TI-30XS MultiView™ calculator, enter the following: $\frac{1}{3} \text{ } 1 \text{ } \downarrow 3 \text{ } \times 40 \text{ } x^2 \text{ } \times 81 \text{ } \text{enter}$.

The right side of the display shows 43200.

The volume of the pyramid is **43,200 cubic feet**.

You may need your calculator to find the exact answer to problems that involve the Pythagorean relationship. These problems can also be done in one series of keystrokes.

Example 2: A right triangle has legs 10 inches and 24 inches in length. What is the length of the hypotenuse?

Use the formula $a^2 + b^2 = c^2$. You will need to add the squares of the legs and then find the square root of the total.

Press: $10 \text{ } x^2 \text{ } + 24 \text{ } x^2 \text{ } \text{enter}$. The display will read 676. Then press $\text{2nd } x^2$ 676 enter . The right side of the display shows 26.

The hypotenuse is **26 inches** in length.

If you need to include the quantity pi (π) in a calculation, the TI-30XS MultiView™ calculator includes a π key, located three keys below the 2nd key on the left-hand side of the keypad.

Example 3: What is the area of a circle with radius 2.4 cm? Round your answer to the nearest tenth.

Use the formula $\text{Area} = \pi r^2$. Press $\pi \text{ } \times 2.4 \text{ } x^2$ and then press enter . The right side of the display reads 18.09557368.

Round that to find the answer: **18.1 cm²**.

GED® TEST TIP

Watch the calculator display while you are entering all of these keystrokes. If you make a mistake, use the delete key to remove that mistake and then replace it with the correct number or function. Practice that with these examples.

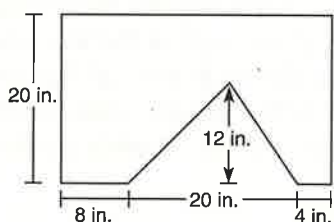
Practice 9

A. Use your calculator to evaluate each formula.

- Find the perimeter of a rectangle with a length of 16 inches and a width of 5 inches.
Perimeter = $2 \times \text{length} + 2 \times \text{width}$
- Find the area of a triangle with a base of 26 centimeters and a height of 15 centimeters.
Area = $\frac{1}{2} \times \text{base} \times \text{height}$
- What is the volume of a cube if the edge measures 3.5 feet? Round to the nearest cubic foot.
Volume = edge^3
- What is the measure of the hypotenuse of a right triangle when the legs measure 13 cm and 9 cm? Round your answer to the nearest tenth.
Pythagorean relationship: $a^2 + b^2 = c^2$
- Find the circumference of a circle with a diameter of 12 inches. Round to the nearest tenth.
Circumference = $\pi \times \text{diameter}$
- Find the volume of a cone with a radius of 12 cm and a height of 20 cm. Round to the nearest cm^3 .
Volume = $\frac{1}{3} \times \pi \times \text{radius}^2 \times \text{height}$

B. Choose the one best answer to each question. You MAY use your calculator.

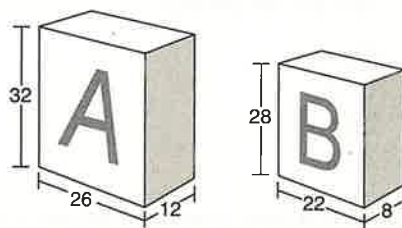
Question 7 refers to the following drawing.



- For a woodworking project, Paul cuts the shape shown above from plywood. What is the area in square inches of the piece? (Hint: Think of the shape as a triangle removed from a rectangle.)
A. 400
B. 480
C. 520
D. 640

- To the nearest cubic meter, what is the volume of a cylinder with a radius of 1.5 meters and a height of 5 meters?
A. 25
B. 35
C. 45
D. 141

Questions 9 and 10 refer to the drawing.



All measurements are in centimeters.

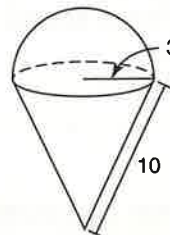
- How many cubic centimeters greater is the volume of Box A than the volume of Box B?
A. 64
B. 1,448
C. 4,928
D. 5,056
- An advertiser plans to print advertisements on one side panel of each of the boxes (the shaded faces in the drawing). What is the total area, in square centimeters, that the advertiser will cover?
A. 224
B. 384
C. 608
D. 1,448

Answers and explanations begin on page 684.

Geometry Practice Questions

- What is the surface area, in square centimeters, of a cube with each edge length of 5 cm?
 - 15
 - 25
 - 125
 - 150
- What is the surface area, in square centimeters, of a cylinder with a height of 10 cm and a radius of 6 cm?
 - 136π
 - 160π
 - 192π
 - 256π
- What is the surface area, in square centimeters, of a cylinder that measures 8 cm tall with a diameter of 6 cm on its base?
 - 48π
 - 57π
 - 66π
 - 132π
- A pyramid has a square base. If each edge of the pyramid is 4, what is the total surface area of the entire pyramid?
 - $8\sqrt{3}$
 - $8\sqrt{3} + 16$
 - $16\sqrt{3} + 16$
 - $64\sqrt{3}$
- The floor of a walk-in closet measures 7 feet by 4 feet. If the ceiling height is 8 feet, what is the volume in cubic feet of the closet?
 - 28
 - 56
 - 112
 - 224

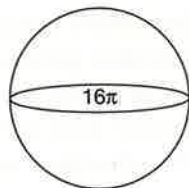
Question 7 refers to the following figure.



- What is the surface area, in square centimeters, of a cylinder that measures 8 cm tall with a diameter of 6 cm on its base?
 - 48π
 - 57π
 - 66π
 - 132π
- A pyramid has a square base. If each edge of the pyramid is 4, what is the total surface area of the entire pyramid?
 - $8\sqrt{3}$
 - $8\sqrt{3} + 16$
 - $16\sqrt{3} + 16$
 - $64\sqrt{3}$
- An ice cream cone has a spherical scoop of ice cream placed inside. Exactly one-half of the ice cream is visible above the rim of the cone. What is the surface area of the entire ice cream treat, in units squared?
 - 24π
 - 30π
 - 48π
 - 60π
- The radius of Sphere A is 2, and the radius of Sphere B is 1. What is the ratio of the volume of Sphere A to that of Sphere B? (*Hint: Use the formula for the volume of a sphere on page 398.*)
 - 1:2
 - 4:1
 - 8:1
 - 8:3
- A parallelogram has a base of 6 centimeters and a height of 10 centimeters. This parallelogram has the same area as a triangle with a height of 5 centimeters. What is the measure, in centimeters, of the base of the triangle?

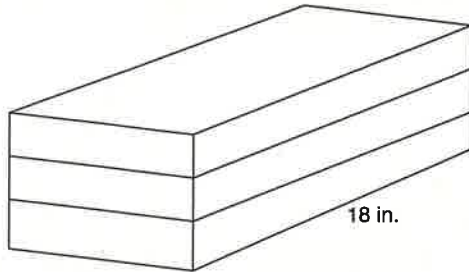
Write your answer in the box.

Question 5 refers to the following figure.



- A toy factory paints all of its rubber balls with 2 coats of latex for durability. How many square centimeters of latex are needed to cover a rubber ball with a circumference of 16π cm?
 - 64π
 - 128π
 - 256π
 - 512π

Question 10 refers to the following figure.



10. Three identical rectangular boxes are stacked one on top of another. The total height of all three boxes is the same as the width of one box. The length of the boxes is 18 in., and the width is half the length. What is the combined volume, in cubic inches, of all three boxes?

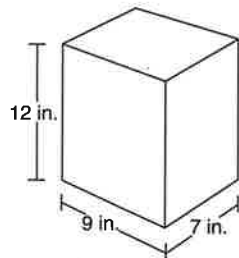
A. 1458
B. 1800
C. 2124
D. 3136

11. Using a compass, Max hikes 300 yards due north of his campsite. From that point, he hikes 400 yards due east. If he were to hike directly to his campsite from this point, how many yards would he have to hike?

A. 400
B. 500
C. 600
D. 700

12. The rectangular base of a container is 9 inches long and 7 inches wide. By how many cubic inches will the volume of the container increase if you increase the length of the base by 2 inches?

A. 84
B. 126
C. 168
D. 216



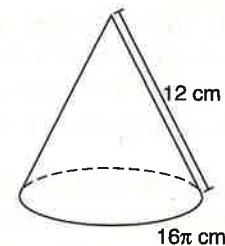
13. In a right triangle, the hypotenuse measures 15 inches. If one leg of the triangle measures 6 inches, which of the following equations could be used to find the length of the other leg (x) in inches?

A. $x = \sqrt{15 - 6}$
B. $x = 15 - 6$
C. $x^2 = 15^2 + 6^2$
D. $x^2 = 15^2 - 6^2$

14. The length of a rectangle is three times its width. If the perimeter of the rectangle is 96 inches, what is its length in inches?

A. 12
B. 24
C. 36
D. 48

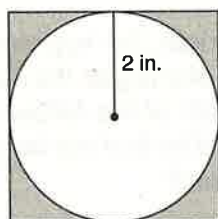
Question 15 refers to the following figure.



15. Sarah is making party hats for a birthday party. Each hat is in the shape of a cone and will have a circumference of 16π cm and a slant height of 7 cm. She will be making 20 hats. What is the total surface area, in square centimeters, of all 20 hats combined?

A. 54π
B. 96π
C. 192π
D. $1,920\pi$

Question 16 refers to the following figure.



16. Use your calculator to make the following calculation. In the figure above, a square contains a circle such that each of the sides of the square is touching a point on the circle. What is the approximate area, in square inches, of the shaded portion?

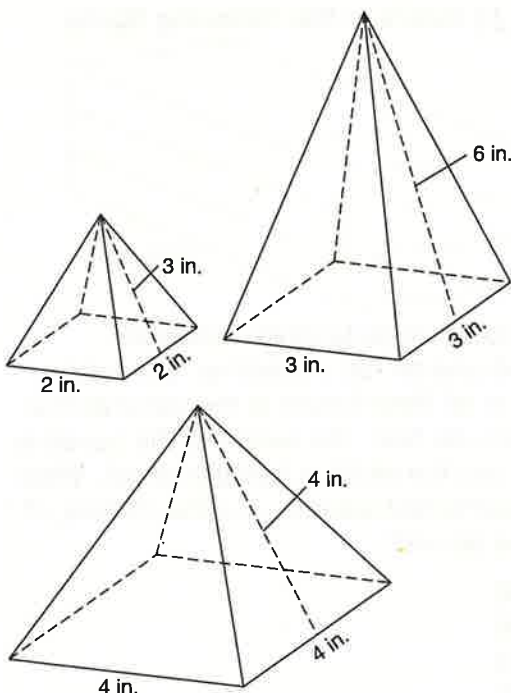
A. 3.43
B. 12.57
C. 16.00
D. 28.57

Questions 17 and 18 refer to the following.

A classroom is 40 feet long and 25 feet wide. The ceiling height is 12 feet. The school district plans to repaint the room and put in air-conditioning. The ceiling tile will not be painted.

17. What is the approximate total square footage of the four walls of the room? Ignore space taken up by windows and doors.
- A. 1200
B. 1560
C. 1920
D. 4000
18. To choose an air-conditioning system, the school district must know the volume of the room. What is the volume in cubic feet?
- A. 1000
B. 3120
C. 12000
D. 15625

Question 19 refers to the following figures.



19. Laura is painting model pyramids for an art project. She needs one tube of paint for every 30 square inches of surface area. If Laura paints each pyramid and its base, approximately how many tubes of paint should she buy?

A. 4
B. 16
C. 48
D. 109

Answers and explanations begin on page 684.

GED® Test Mathematics Formula Sheet

Area of a:

square	$A = s^2$
rectangle	$A = lw$
parallelogram	$A = bh$
triangle	$A = \frac{1}{2}bh$
trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$
circle	$A = \pi r^2$

Perimeter of a:

square	$P = 4s$
rectangle	$P = 2l + 2w$
triangle	$P = s_1 + s_2 + s_3$
Circumference of a circle	$C = 2\pi r$ OR $C = \pi d$; $\pi \approx 3.14$

Surface area and volume of a:

rectangular/right prism	$SA = ph + 2B$	$V = Bh$
cylinder	$SA = 2\pi rh + 2\pi r^2$	$V = \pi r^2 h$
pyramid	$SA = \frac{1}{2}ps + B$	$V = \frac{1}{3}Bh$
cone	$SA = \pi rs + \pi r^2$	$V = \frac{1}{3}\pi r^2 h$
sphere	$SA = 4\pi r^2$	$V = \frac{4}{3}\pi r^3$

(p = perimeter of base with area B ; $\pi \approx 3.14$)

Data

mean	mean is equal to the total of the values of a data set, divided by the number of elements in the data set
median	median is the middle value in an odd number of ordered values of a data set, or the mean of the two middle values in an even number of ordered values in a data set

Algebra

slope of a line	$m = \frac{y_2 - y_1}{x_2 - x_1}$
slope-intercept form of the equation of a line	$y = mx + b$
point-slope form of the equation of a line	$y - y_1 = m(x - x_1)$
standard form of a quadratic equation	$y = ax^2 + bx + c$
quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Pythagorean Theorem	$a^2 + b^2 = c^2$
simple interest	$I = Prt$ (I = interest, P = principal, r = rate, t = time)
distance formula	$d = rt$
total cost	total cost = (number of units) $\times$ (price per unit)

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